



Advanced Internet of Things Technology

Week 1: Modern IoT Stack & Digital Twin Architectures

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Course Introduction



Course title and scope

- Advanced Internet of Things Technology (Master's level)
- Focus on architecture, intelligence, and system integration
- Coverage from devices to edge, cloud, and digital twins

Programme and logistics

- Master of Science (M.Sc.) programme
- Academic year and weekly structure
- Combination of theory lectures and project-based learning

Quick survey

<https://www.menti.com/alui193jah18>



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Pedagogical Approach



The vertical slice concept

- One complete IoT system built across the semester
- Hardware/Simulator, networking, intelligence, and application layers
- Continuous refinement rather than disconnected labs

Learning outcomes

- Ability to design scalable IoT architectures
- Understanding of digital twin-enabled systems
- Preparation for research and industry practice

Project Structure Overview



Semester-long capstone project

- Students work in fixed pairs
- One project spanning all technical modules
- Progressive development from simple to advanced systems

Levels of project complexity

- Level 1: connected and observable systems
- Level 2: intelligent edge or enriched digital twin
- Level 3: cognitive systems with AI-driven reasoning

Course Overview



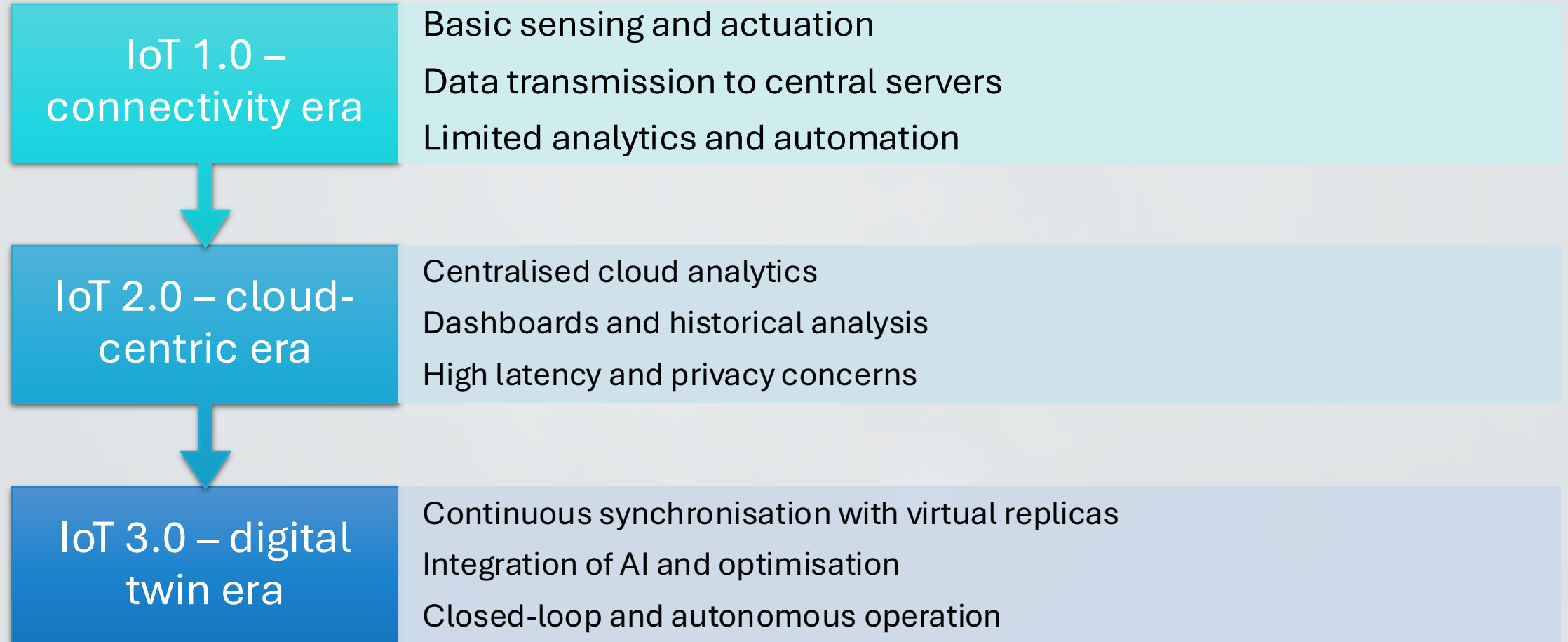
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Week 1 – Modern IoT Stack & Digital Twin Architectures

Evolution of IoT Architectures



The Paradigm Shift: From Connected Devices to Cognitive Systems

IoT 1.0: The Connectivity Era



- Simple remote monitoring and telemetry.
- Focus: 'Does it work?'
- Architecture: Device $\rightarrow$ Central Server.
- Limitation: High latency, data overload, minimal autonomy.

IoT 2.0: The Cloud-Centric Era



- Centralized cloud analytics and dashboards.
- Focus: 'What happened?' (Historical analysis).
- Architecture: Device $\rightarrow$ Cloud Gateway $\rightarrow$ Analytics Platform.
- Limitation: Still reactive, privacy concerns, dependent on connectivity.

IoT 3.0: The Digital Twin Era



- Real-time synchronization with virtual replicas.
- Focus: 'What will happen, and what should we do?' (Prediction & Optimization).
- Architecture: A closed loop between the physical asset and its intelligent twin.
- Benefit: Enables autonomy, low-latency decisions, and cognitive understanding.

The Cognitive Digital Twin: A 3-Level Architecture for Modern IoT

Evolution of IoT systems from simple connected devices to sophisticated, AI-driven cognitive systems.

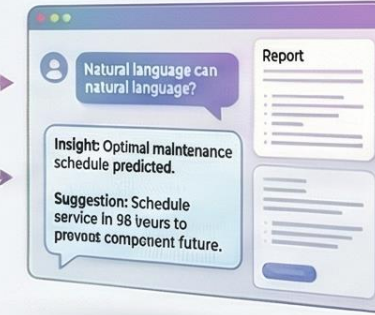
Generative AI Integration:
A Large Language Model (LLM) analyzes data logs to generate reports and advice.



Level 3: The Cognitive System

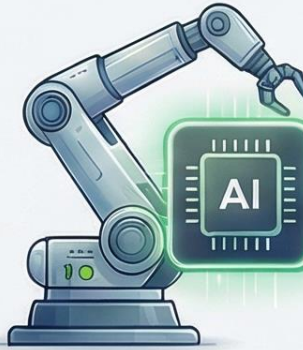
State-of-the-Art Goal: "It Speaks"

The system can interpret data, generate insights in natural language, and suggest actions.



Emergent Autonomy:
The system can automatically suggest optimizations or trigger actions based on its analysis.

On-Device AI (TinyML):
A machine learning model runs directly on the microcontroller to detect anomalies locally.



Level 2: The Intelligent Edge

Intelligence is moved from the cloud to the device itself for faster, more private decisions.



Rich Twin Representation:
The dashboard evolves to a 3D model or map that updates its state in real-time.

Real-Time Data Sync:
Physical sensors stream data to the twin via protocols like MQTT or CoAP.



Level 1: The Connected Twin

A live, synchronized digital replica for basic monitoring and visualization.



Foundational Visualization:
A simple dashboard displays the live state of the physical asset.

Digital Twin vs Related Concepts

Twin, Shadow, and Simulation

Digital Shadow



A digital representation that changes when the physical asset changes, but not vice-versa. Useful for monitoring.

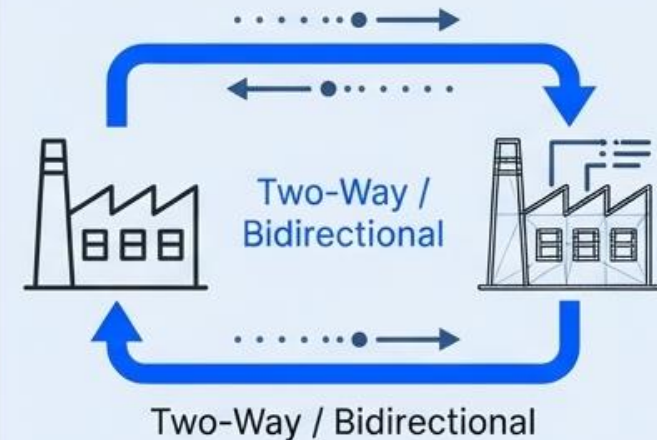
Simulation



Offline / What-If

An offline model that explores potential outcomes. It is not continuously synchronized with a live, real-world asset.

Digital Twin



A live, synchronized virtual replica that enables **closed-loop control**. Changes in the physical world affect the twin, and actions taken on the twin can affect the physical world. This is the foundation of autonomy.

Types of Digital Twins



Structural classification

- Discrete digital twins for single assets
- Composite digital twins for systems of systems
- Hierarchical organisation of twins

Functional classification

- Product-level digital twins
- Facility-level digital twins
- Lifecycle-oriented digital twins

Core Characteristics of Digital Twins



Network Connectivity

The bidirectional link between the physical asset and its twin.



Synchronization

The process ensuring the virtual replica accurately mirrors the physical state.



Fidelity

The accuracy and precision of the digital model. Assessed by the frequency and detail of updates.



Twinning Rate

The defined frequency of data exchange (e.g., every 100ms, every 5 seconds).



Persistency

The twin remains available and stores data throughout the asset's lifecycle, even if the physical object is offline or destroyed.



Composability

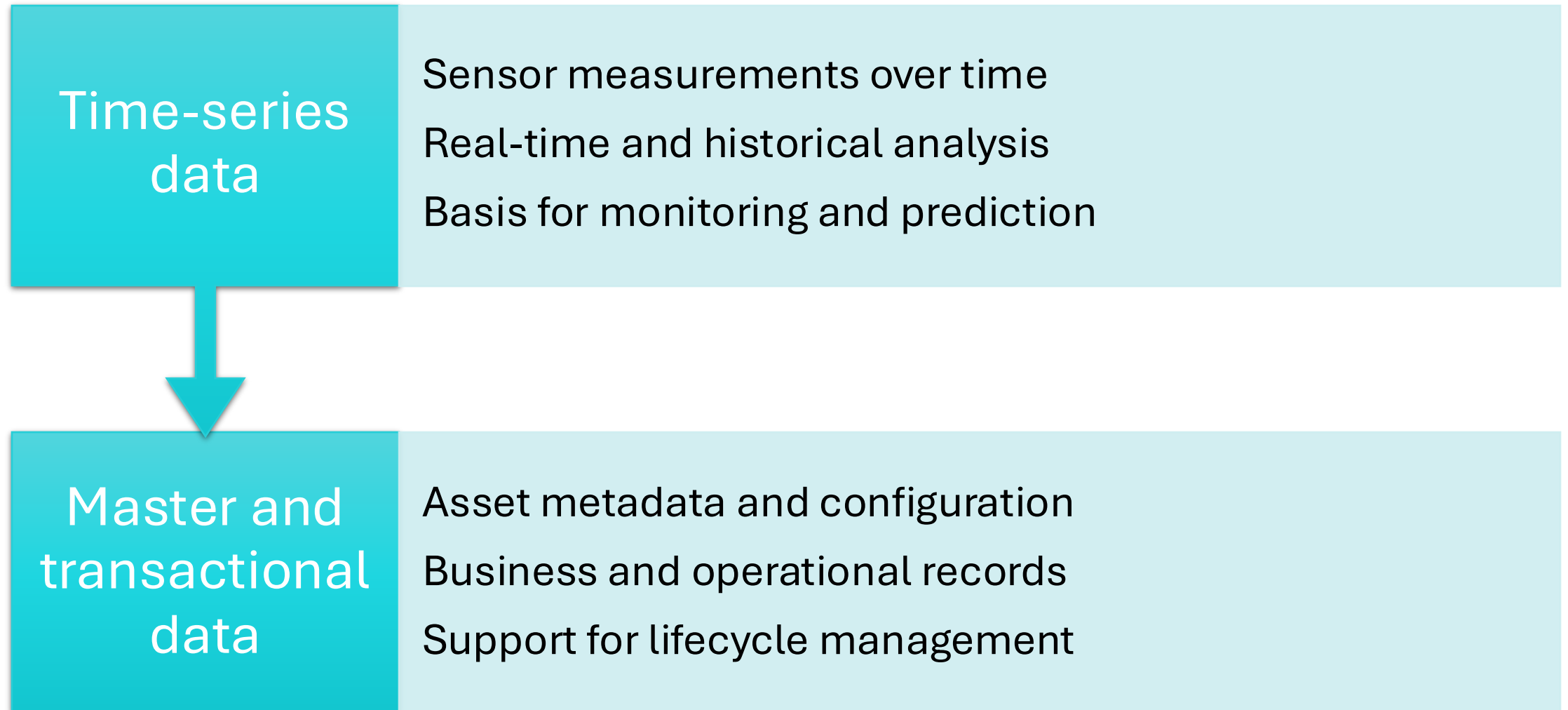
The ability to aggregate multiple discrete twins to represent a complex system of systems (e.g., twinning a factory from machine twins).



Contextualization

The twin understands its surrounding environment and the context of the data it receives.

Data in Digital Twins



"How do we describe a device?"

The Old Way: Random JSON blobs.

The New Way (Standards):

AAS (Asset Administration Shell): The Industry 4.0 standard.

DTDL (Digital Twin Definition Language): Microsoft/Azure standard.

NGSI-LD: ETSI standard for Smart Cities (Linked Data).

Digital Twin Definition Language (DTDl)



<https://learn.microsoft.com/en-us/azure/digital-twins/concepts-models>

<https://github.com/Azure/opendigitaltwins-dtdl/blob/master/DTDl/v3/DTDl.v3.md>

Models in Digital Twins

Physics-based models

- Representation using physical laws
- High accuracy for engineering systems
- Computationally intensive simulations

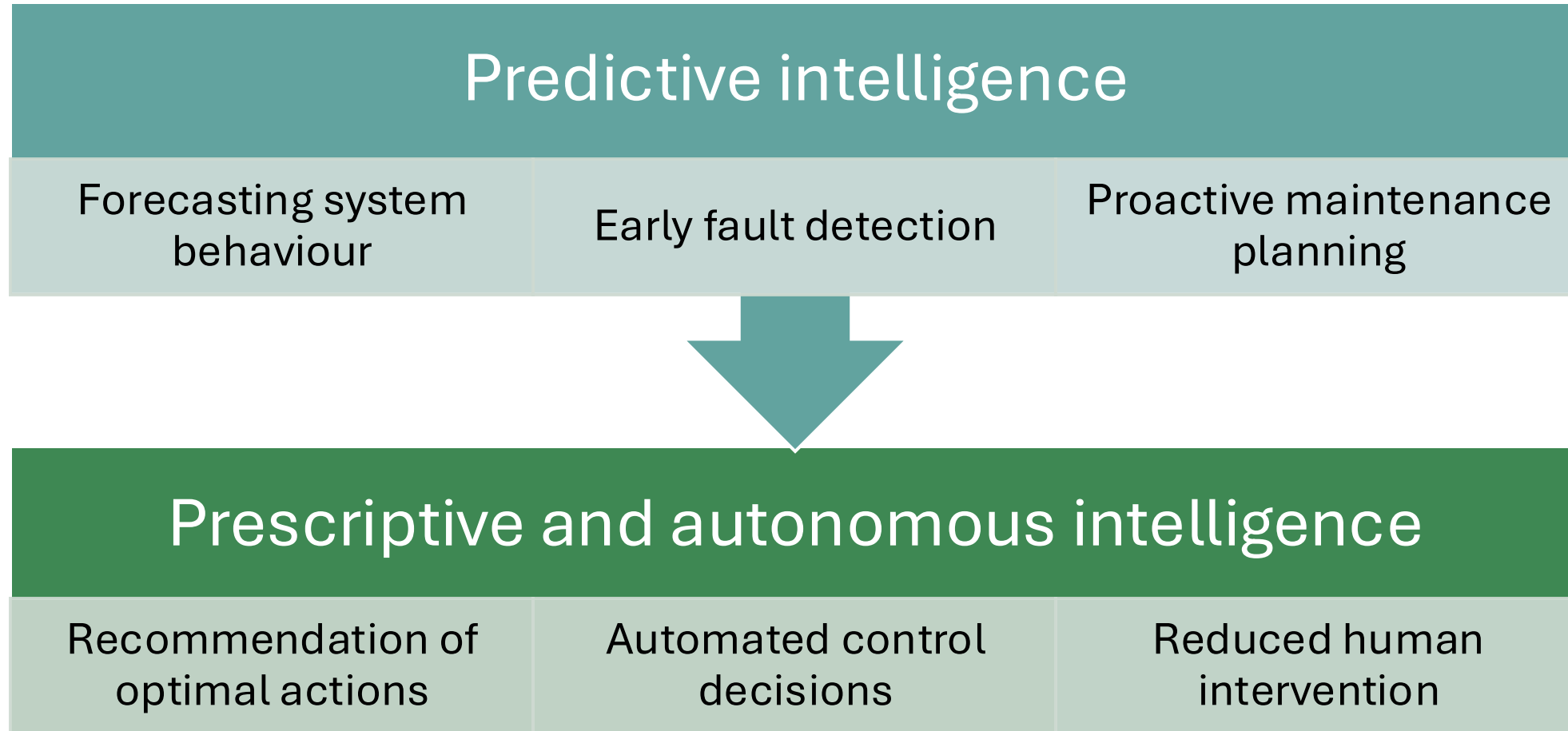
Analytics and AI models

- Machine learning-based prediction
- Pattern recognition and anomaly detection
- Adaptive system behaviour

Visual and simulation models

- 3D models and virtual environments
- Augmented and virtual reality
- Improved human understanding

Role of AI in Digital Twins



Intelligence Placement



Cloud intelligence

- High computational power
- Global system visibility
- Suitable for long-term analytics

Edge intelligence

- Low-latency decision making
- Reduced bandwidth usage
- Improved data privacy

Hybrid intelligence

- Collaboration between edge and cloud
- Balanced latency and scalability
- Practical design choice for most systems

Checkpoint

1. Which digital twin layer seems most complex to you?
2. How important is interoperability in real deployments?
3. Where should intelligence be placed in the architecture?

Project Logistics



Group formation

- Two students per group
- Complementary technical skills
- Long-term collaboration

Early project expectations

- Identify application domain
- Think in terms of system architecture
- Prepare for Level 1 implementation

Capstone Project Menu: Choose Your Vertical

Select one topic. Your proposal must define how you will achieve Level 1, 2, and 3 within this vertical.

Industry 4.0

- **The Self-Diagnosing Motor**
(Vibration analysis + Maintenance Chatbot) 
- **Visual Quality Control**
(Defect detection + Report Generator) 
- **Predictive Maintenance Rig**
(Remaining Useful Life of a fan/pump) 

Smart Cities

- **Intelligent Traffic Intersection**
(Car counting + Timing Optimizer) 
- **Structural Health Monitor**
(Bridge vibration + Integrity Report) 
- **Acoustic Incident Monitor**
(Gunshot detection + Dispatch Summarizer) 

Internet of Medical Things (IoMT)

- **Elderly Fall Guardian**
(IMU-based fall detection + Emergency Call) 
- **Remote Cardiac Monitor**
(Arrhythmia classification + Health Summary) 
- **Smart Physio Glove**
(Gesture tracking for rehab + Motivational Coach) 

Smart Home (Matter Protocol)

- **Energy Disaggregation Meter**
(Appliance identification + Cost Advisor) 
- **Gesture Control Hub**
(Radar-based gesture recognition + Scene Generator) 
- **Security Sentry**
(Facial Recognition + Visitor Log Summarizer) 

Looking Ahead



Reflection and Preparation

- What is your understanding of a cognitive IoT system?
- Which project ideas are emerging for you?
- What skills should you strengthen before Week 2?

Week 2 preview

- Introduction to edge intelligence
- TinyML and on-device inference
- Motivation for moving intelligence closer to devices

Preparation guidance

- Review basic machine learning concepts
- Reflect on latency and privacy trade-offs
- Think about potential project ideas

Wrap-Up and Discussion



Key takeaways

IoT is evolving toward intelligent systems
Digital twins are central to modern architectures
System thinking is essential



Open discussion

Questions on digital twin concepts
Initial project idea brainstorming
Clarification of course expectations

References

- [1] Hakiri, A., Gokhale, A., Yahia, S.B. and Mellouli, N., 2024. A comprehensive survey on digital twin for future networks and emerging Internet of Things industry. *Computer Networks*, 244, p.110350.
- [2] Xu, H., Wu, J., Pan, Q., Guan, X. and Guizani, M., 2023. A survey on digital twin for industrial internet of things: Applications, technologies and tools. *IEEE Communications Surveys & Tutorials*, 25(4), pp.2569-2598.

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